Karnav Prajapati

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Professional Summary

Aspiring Machine Learning Engineer and B.E. Information Technology student with a strong foundation in Python, data analysis, and machine learning algorithms. Experienced in data preprocessing, feature engineering, and building supervised learning models using Scikit-learn and TensorFlow through academic projects and hackathons. seeking an entry-level ML or Data Science role to apply analytical and problem-solving skills..

Education

- **A D Patel Institute of Technology** — *Expected May 2027*
  - Bachelor of Engineering in Information Technology
- **R.C. Technical Institute, Ahmedabad** — *Completed May 2023*
  - Diploma in Information Technology — 6.94 CGPA

Technical Skills

Languages: Python, SQL
ML/Data Science Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, OpenCV
Deep Learning Frameworks: TensorFlow, Keras
Developer Tools: Git, GitHub, VS Code, Jupyter Notebooks
Web Technologies: HTML, CSS, JavaScript, ReactJS (Foundational)

Projects

- **Facial Emotion Recognition & Personalized Music Recommendation System** — *Python, TensorFlow, Keras, CNN*
  GitHub Repository
  - Engineered a Convolutional Neural Network (CNN) architecture using TensorFlow/Keras and OpenCV to classify facial expressions (e.g., happy, sad, neutral) with high accuracy..
  - Trained on **10,000+ images** using data augmentation to reduce overfitting and improve generalization.
  - Preprocessed images with OpenCV and evaluated performance using confusion matrices and classification reports.

Automated Program Repair with AST Analysis — **Python, JavaParser, Defects4J**
GitHub Repository
  - Developed research pipeline for Automated Program Repair using Abstract Syntax Tree analysis on Defects4J benchmark (835+ Java bugs)
  - Implemented k-hop neighborhood extraction algorithm with 50-token constraints for minimal-context repair evaluation
  - Analyzed repair patterns to contribute to token-constrained code generation research

Achievements & Competitions

- **Smart India Hackathon (SIH) Finalist** (2024, 2025): Advanced to internal college finals for innovative solution design
- **NEXOTHON 2025 Participant**: 24-hour hackathon focused on rapid prototyping and collaborative development
- **Odoo x Mindbend Hackathon 2025 Participant**: Competitive hackathon at SVNIT Surat for innovative solutions

Certifications

- Python Language by init Labs — *Udemy*
- Machine Learning with Pearson — *Pearson*